Experimental methods in solid-state physics

07 Raman and Brillouin light scattering

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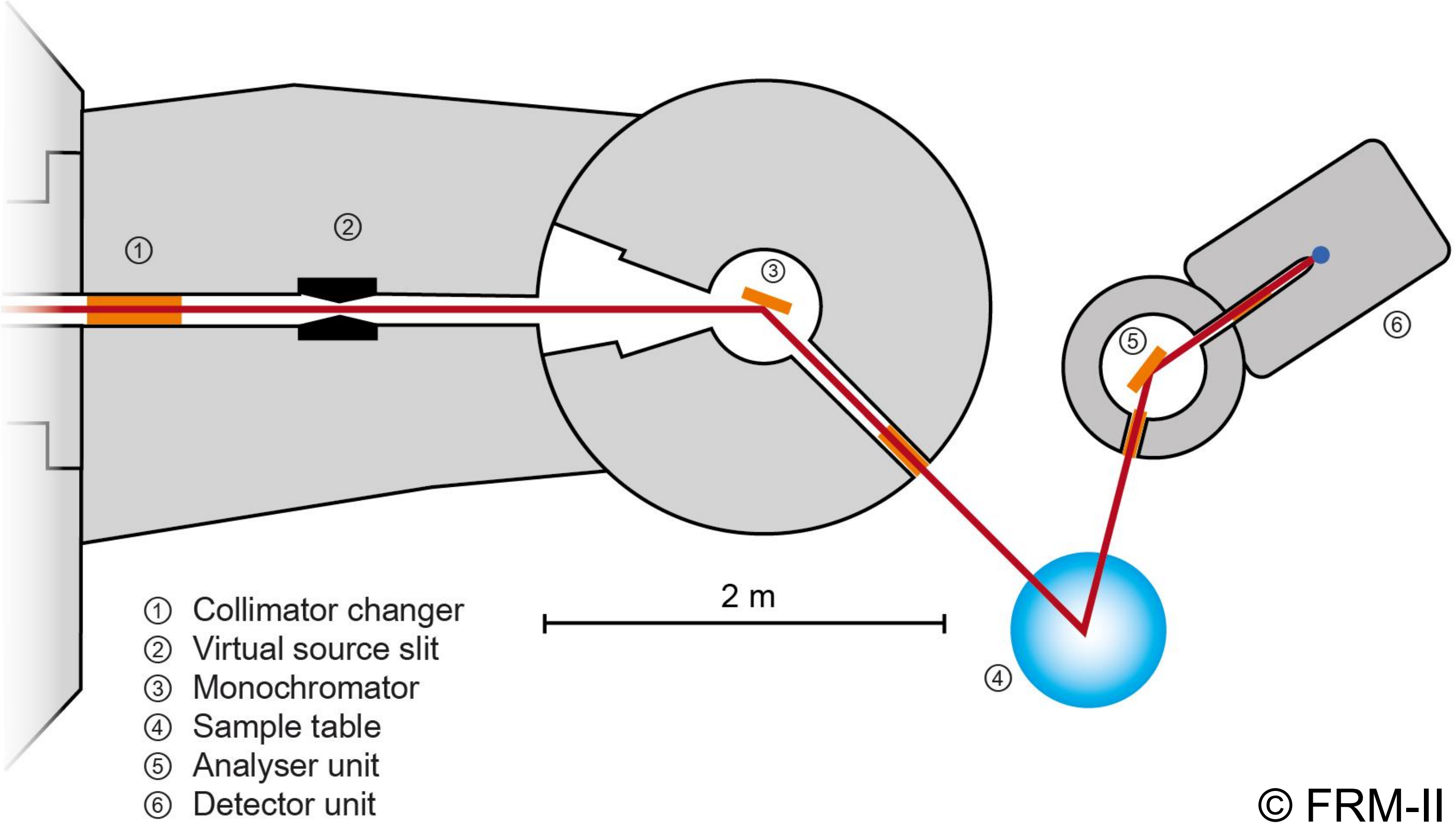
Triple axes spectrometer



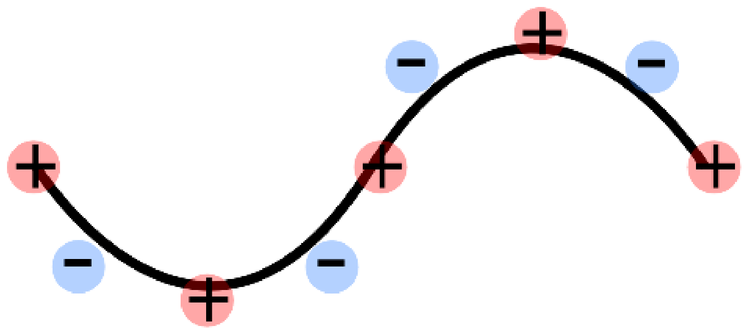
Inelastic scattering

$$q = k - k'$$

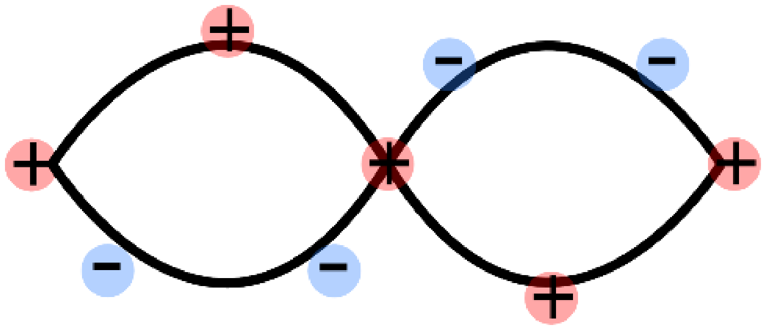
$$\Delta E = \hbar\omega = E - E'$$

Phonon dispersion of GaAs

Acoustical Mode → sound wave



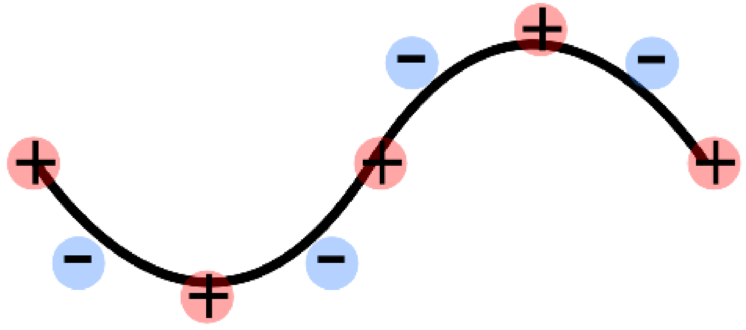
Optical Mode → interacts with light



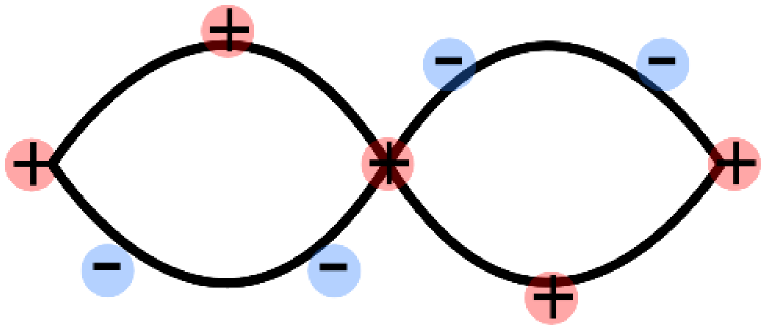
Number of **phonon branches**: 3N
N: number of atoms in the primitive unit cell

Phonon dispersion of GaAs

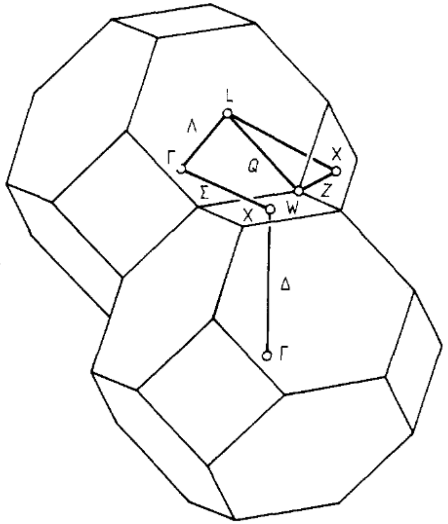
Acoustical Mode → sound wave

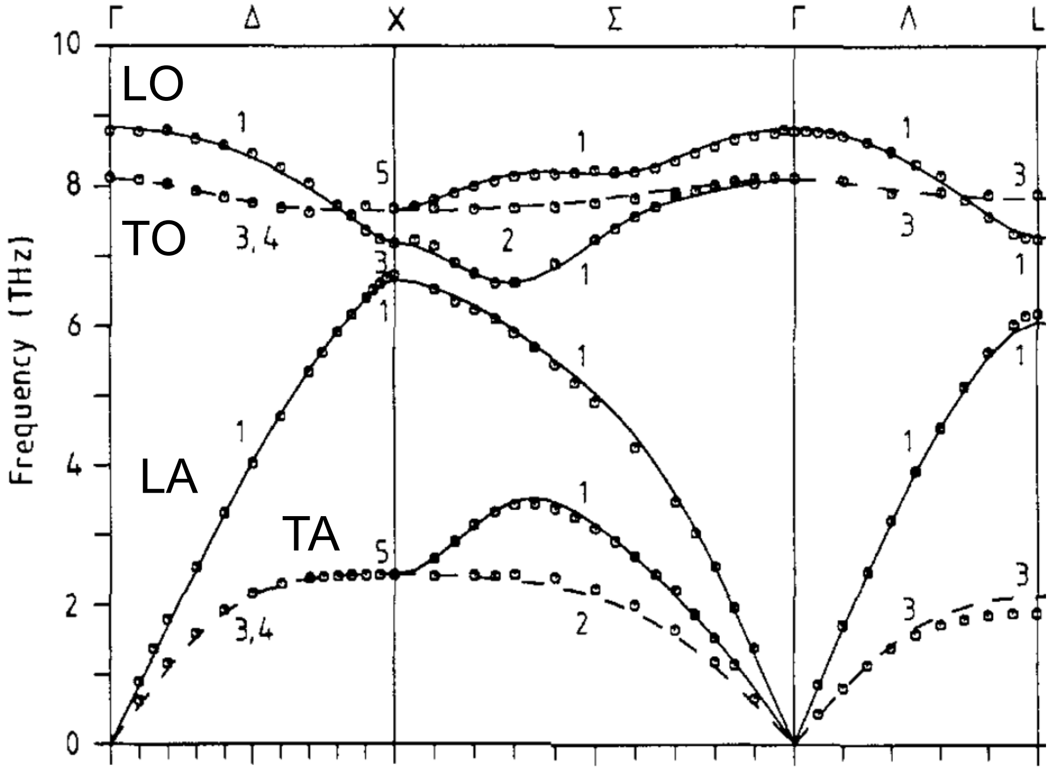


Optical Mode → interacts with light

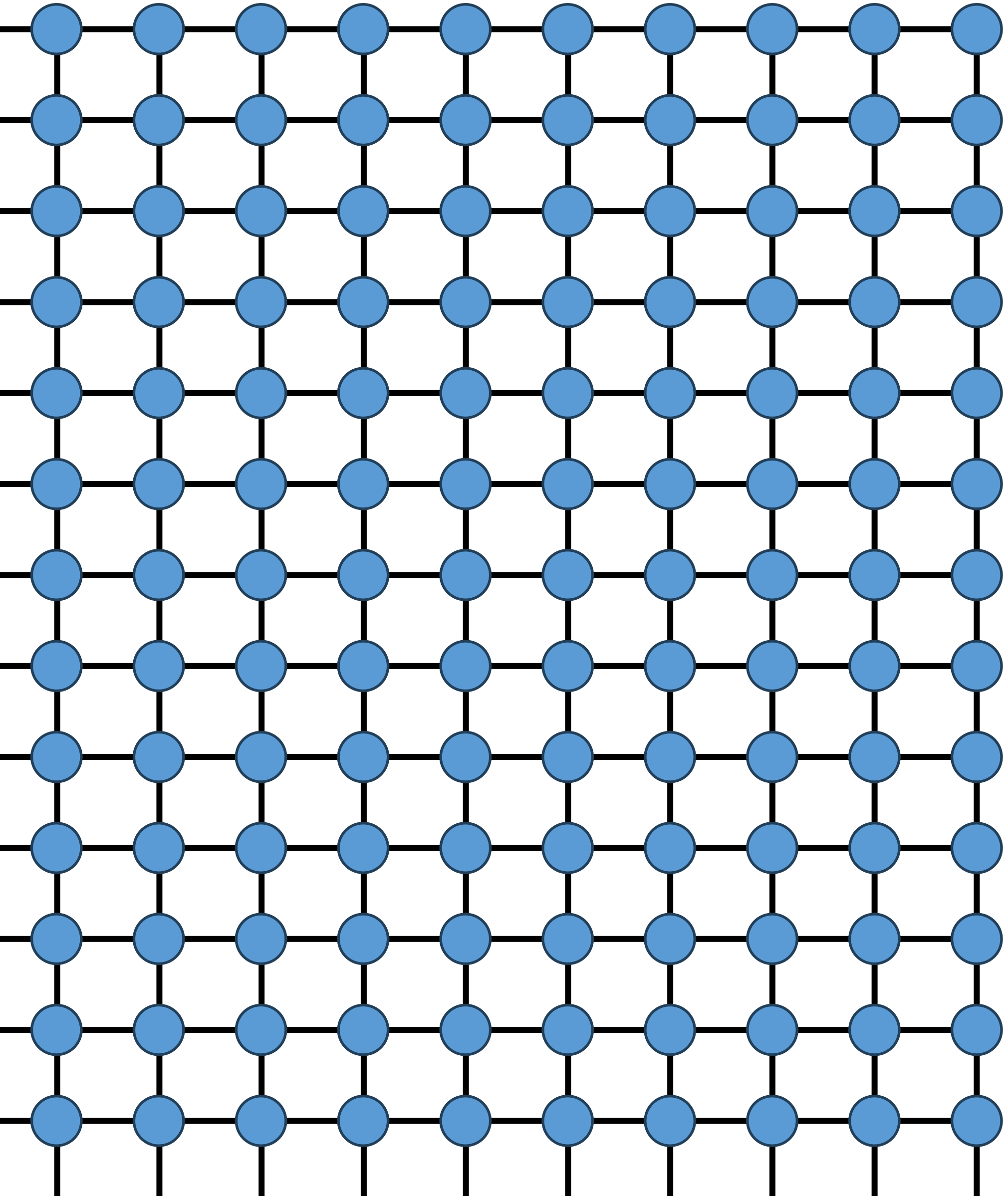


Zinc-blende structure



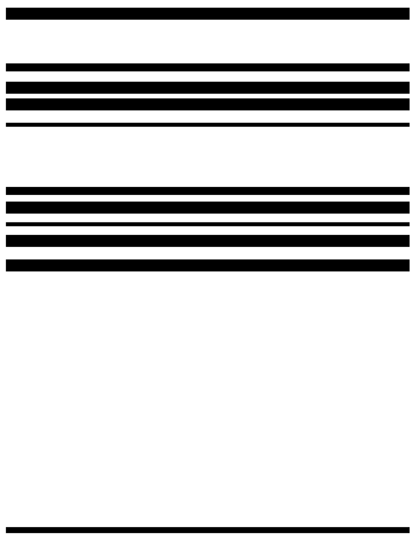


Phonons are quasi-particles

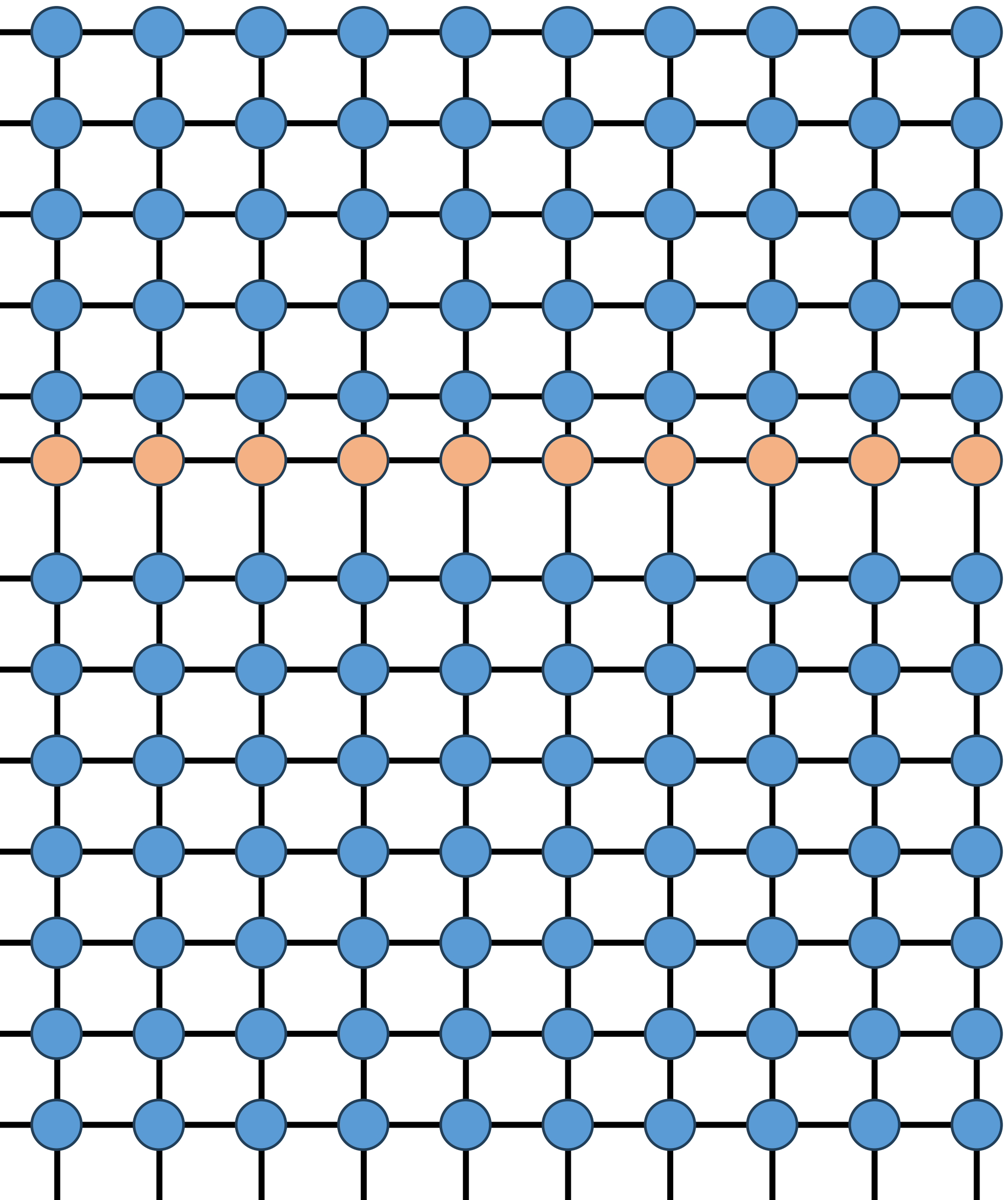


Macroscopic periodic crystal:

- A strongly interacting many-body system
- With a well-defined many-body ground state
- And a large spectrum of possible excited many-body states



Phonons are quasi-particles



Quasiparticles:

•Are *elementary excitations* of that many-body ground state

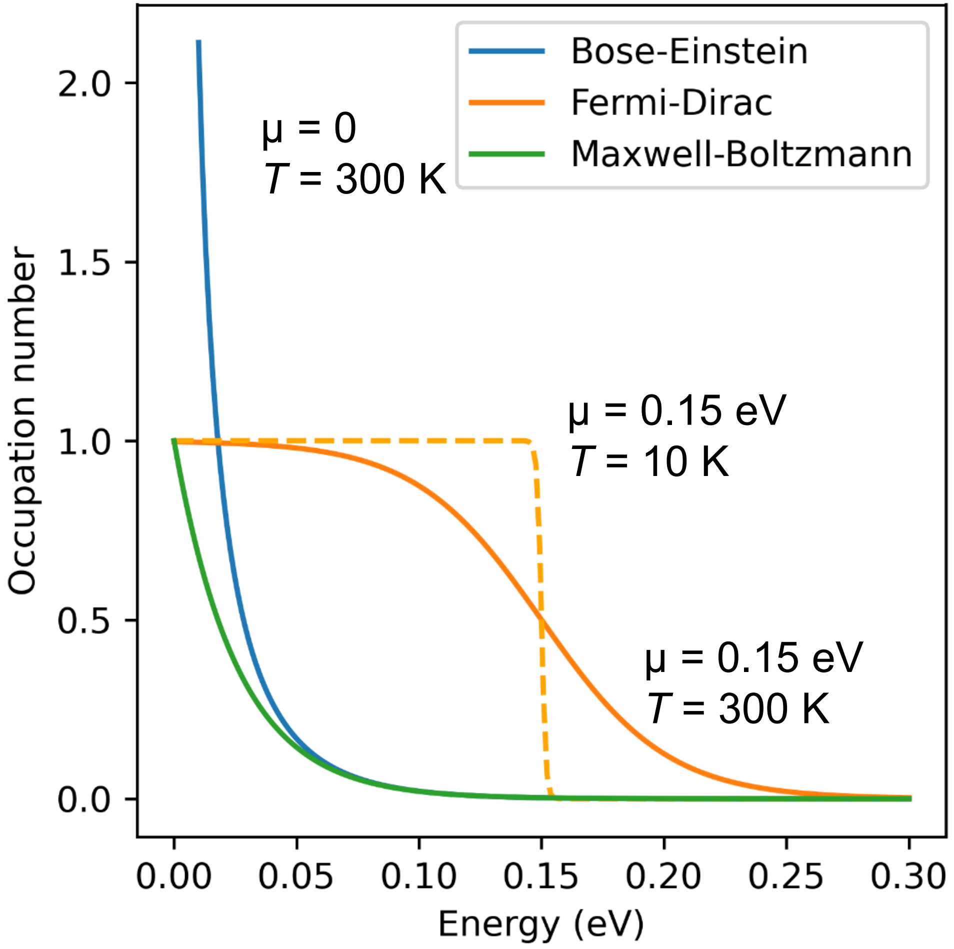
•Behave as if they were particles with effective properties (mass, charge, spin, etc.)

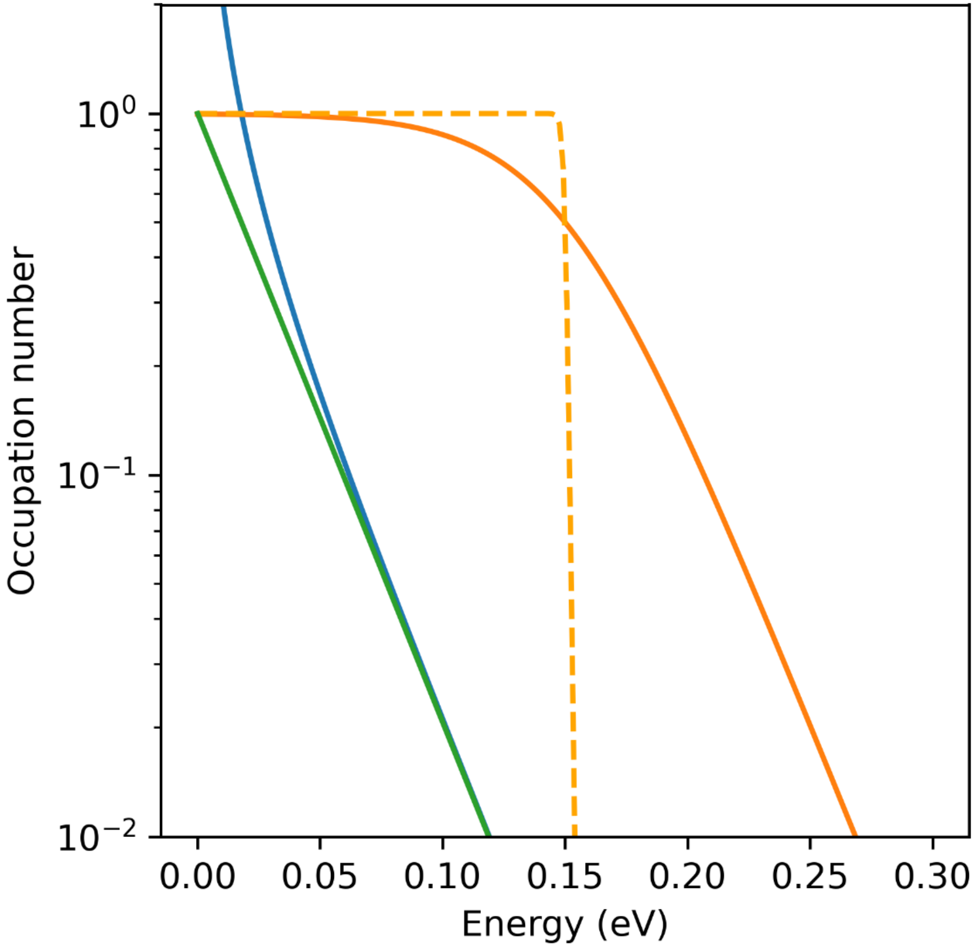
•Represent *collective* motion of many underlying microscopic degrees of freedom

•Have **long lifetimes** (their spectral peak is sharp), meaning interaction/scattering is weak enough to give them well-defined energy $\hbar\omega$ and momentum $\boldsymbol{q}$

Statistics for (quasi-)particles

	Classical (Maxwell–Boltzmann)	**Bosons (Bose–Einstein)**	**Fermions (Fermi–Dirac)**
Distribution function	$\frac{1}{e^{(E-\mu)/(k_B T)}}$	$\frac{1}{e^{(E-\mu)/(k_B T)} - 1}$	$\frac{1}{e^{(E-\mu)/(k_B T)} + 1}$



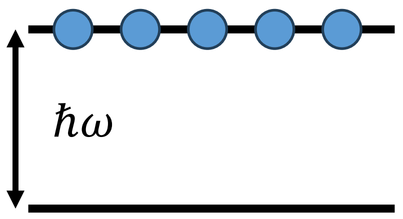


Statistics for (quasi-)particles

	Classical (Maxwell–Boltzmann)	Bosons (Bose–Einstein)	Fermions (Fermi–Dirac)
Indistinguishability	*Not enforced* (particles treated as distinguishable)	Enforced	Enforced
Quantum state occupancy	No restriction	**Unlimited** per state	**≤ 1** per state (Pauli principle)
Distribution function	$\frac{1}{e^{(E-\mu)/(k_B T)}}$	$\frac{1}{e^{(E-\mu)/(k_B T)} - 1}$	$\frac{1}{e^{(E-\mu)/(k_B T)} + 1}$
Validity regime	High (T), low density	All (T)	All (T)
Chemical potential (\mu)	$\mu < 0$, large negative	$\mu < 0$ (→0 at condensation)	$\mu \geq 0$
Many-particle behavior	No collective quantum effects	**Bose condensation**, coherence, superfluidity	**Fermi surfaces**
Examples	Classical gas, dilute ions	Phonons, photons, magnons, excitons, …	Electrons
Low-T limit	Particles freeze out	Macroscopic occupancy of lowest state	Sharp Fermi surface , $f(E) = \delta(E-\mu)$

Occupation of phonon modes

Phonon mode with q and ω



Ground state

A single lattice vibration mode of frequency ω can carry an integer number $n = 0, 1, 2, \ldots$ of phonons. Each phonon has energy $\varepsilon = \hbar\omega$, so that the total energy of the mode is

$$E_n = n\,\hbar\omega.$$

(For clarity we omit the constant zero-point energy $\frac{1}{2}\hbar\omega$, which cancels from all probability ratios.)

Thermal equilibrium. In the canonical ensemble, the probability of occupying a state of energy E_n is proportional to the Boltzmann factor,

$$P_n \propto e^{-E_n/(k_B T)} = e^{-n\hbar\omega/(k_B T)}.$$

Introduce the shorthand

$$x := e^{-\hbar\omega/(k_B T)},$$

The properly normalized probability distribution is therefore

$$P_n = \frac{x^n}{\sum_{m=0}^{\infty} x^m},$$

The normalization factor is the geometric series

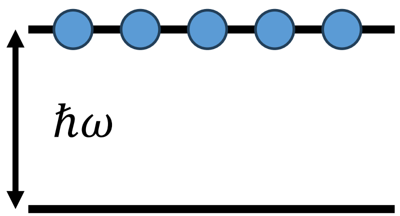
$$Z = \sum_{m=0}^{\infty} x^m = \frac{1}{1-x},$$

which yields

$$P_n = (1-x)\,x^n.$$

Occupation of phonon modes

Phonon mode with q and ω



Ground state

Average phonon occupation. The thermal average of the phonon number is

$$\langle n\rangle = \sum_{n=0}^{\infty} n\,P_n = (1-x)\sum_{n=0}^{\infty} n x^n.$$

Using the identity

$$\sum_{n=0}^{\infty} n x^n = \frac{x}{(1-x)^2},$$

we obtain

$$\langle n\rangle = \frac{x}{1-x} = \frac{1}{e^{\hbar\omega/(k_B T)}-1}.$$

Interpretation. The result

$$\boxed{\langle n\rangle = \frac{1}{e^{\hbar\omega/(k_B T)}-1}}$$

is the Bose–Einstein distribution for a single phonon mode. It follows only from:

- energy quantization in steps of $\hbar\omega$,
- the Boltzmann weight $e^{-E_n/(k_B T)}$,
- and the fact that any integer number of phonons is allowed in the same state with energy (bosons have no exclusion principle).

Light-matter interaction

Minimal coupling. An electron in an electromagnetic field is described by the Hamiltonian

$$\hat{H} = \frac{1}{2m}\big(\hat{\mathbf{p}} + e\mathbf{A}(\mathbf{r},t)\big)^2 + V(\mathbf{r}),$$

written here in Coulomb gauge ($\nabla\cdot\mathbf{A} = 0$, $\phi = 0$). The interaction with light originates from the cross term

$$\hat{H}_{\text{int}} = \frac{e}{2m}\left(\hat{\mathbf{p}}\cdot\mathbf{A}(\mathbf{r},t) + \mathbf{A}(\mathbf{r},t)\cdot\hat{\mathbf{p}}\right)$$

plus a usually negligible A^2 term.

Dipole approximation. For optical wavelengths ($\lambda \sim 500$ nm) the spatial variation of the vector potential is tiny on the scale of atomic wave functions ($r \sim 0.1$ nm). Thus the electromagnetic field is effectively uniform across the unit cell:

$$\mathbf{A}(\mathbf{r},t) \approx \mathbf{A}(t).$$

Under this assumption, $\mathbf{A}(t)$ commutes with $\hat{\mathbf{p}}$, and the interaction term simplifies to

$$\hat{H}_{\text{int}}(t) = \frac{e}{m}\,\hat{\mathbf{p}}\cdot\mathbf{A}(t).$$

This form is often called the *velocity gauge*.

Transformation to the dipole (length) gauge. A time-dependent gauge transformation (see quantum mechanics textbooks) can be used to eliminate the vector potential from the kinetic term when $\mathbf{A}$ is spatially uniform. The transformed Hamiltonian becomes

$$\hat{H} = \hat{H}_0 - \hat{\mathbf{d}} \cdot \mathbf{E}(t), \qquad \hat{\mathbf{d}} = -e\,\hat{\mathbf{r}},$$

with $\mathbf{E}(t) = -\partial \mathbf{A}/\partial t$. This is the *dipole Hamiltonian*. It contains the same physics as the minimal-coupling form but makes the interaction with light explicit: the electric field couples directly to the dipole moment of the electrons.

Connection to polarizability. The expectation value of the dipole operator defines the macroscopic polarization,

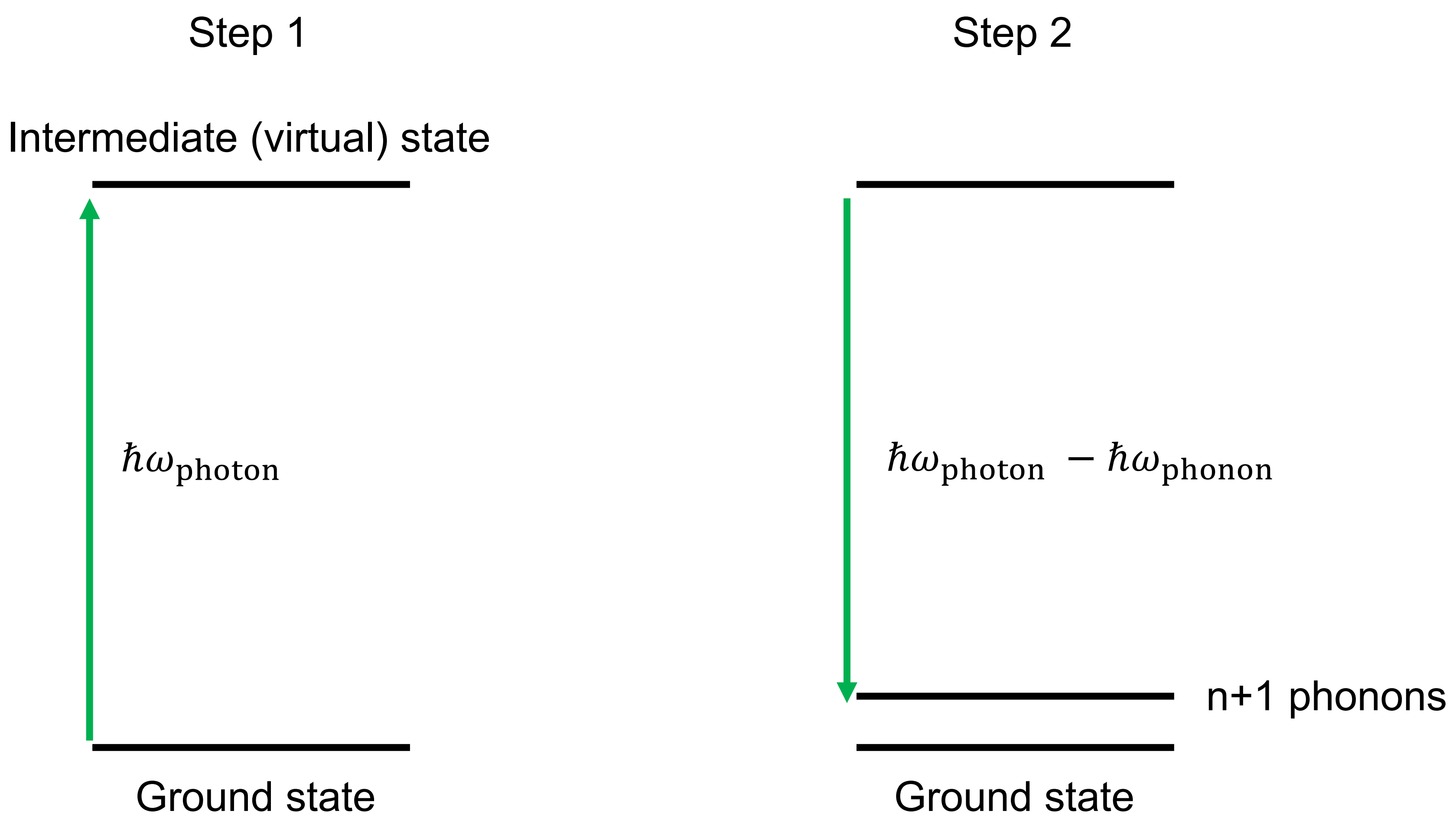
$$\mathbf{P}(t) = \langle \hat{\mathbf{d}}(t) \rangle,$$

which for weak fields responds linearly to $\mathbf{E}(t)$:

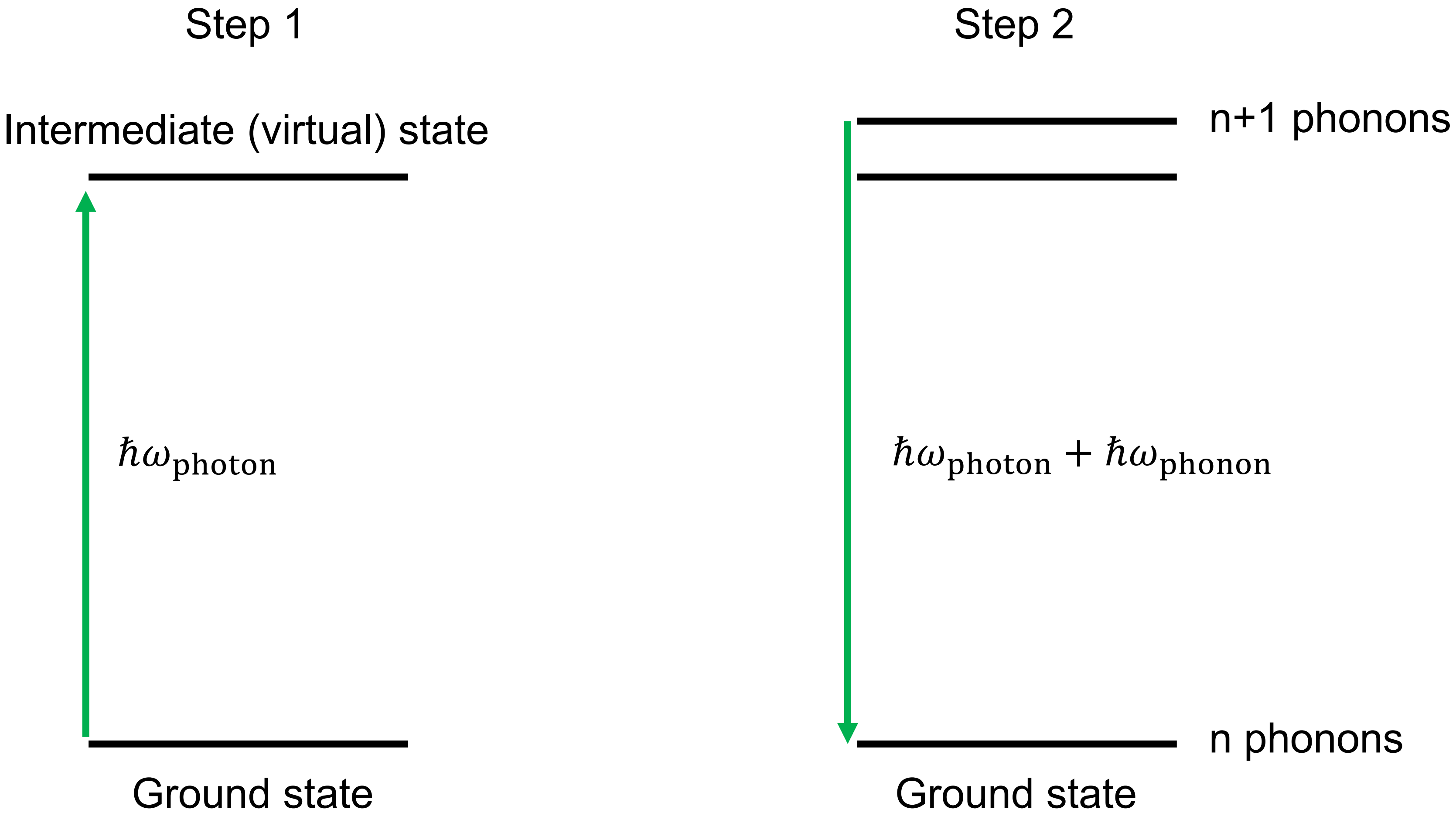
$$P_i(\omega) = \sum_j \alpha_{ij}(\omega)\, E_j(\omega).$$

The tensor $\boldsymbol{\alpha}(\omega)$ is the *polarizability*, whose derivatives with respect to vibrational coordinates generate inelastic light-scattering processes. In particular, a phonon-induced modulation of $\boldsymbol{\alpha}$ produces Raman and Brillouin sidebands in the scattered light.

Raman scattering is a two-step process (Stokes)



Raman scattering is a two-step process (anti-Stokes)



Raman scattering is a two-step process

Dynamical structure factor: which excitation exist in the system

$$S(\mathbf{q},\omega) \;\propto\; \sum_s |\mathbf{e}_{\mathbf{q}s}\cdot\hat{\epsilon}|^2 \left[n(\omega_{\mathbf{q}s})\,\delta(\omega+\omega_{\mathbf{q}s}) + \big(n(\omega_{\mathbf{q}s})+1\big)\,\delta(\omega-\omega_{\mathbf{q}s})\right]$$

Raman vertex (transition matrix element): which transitions are optically active

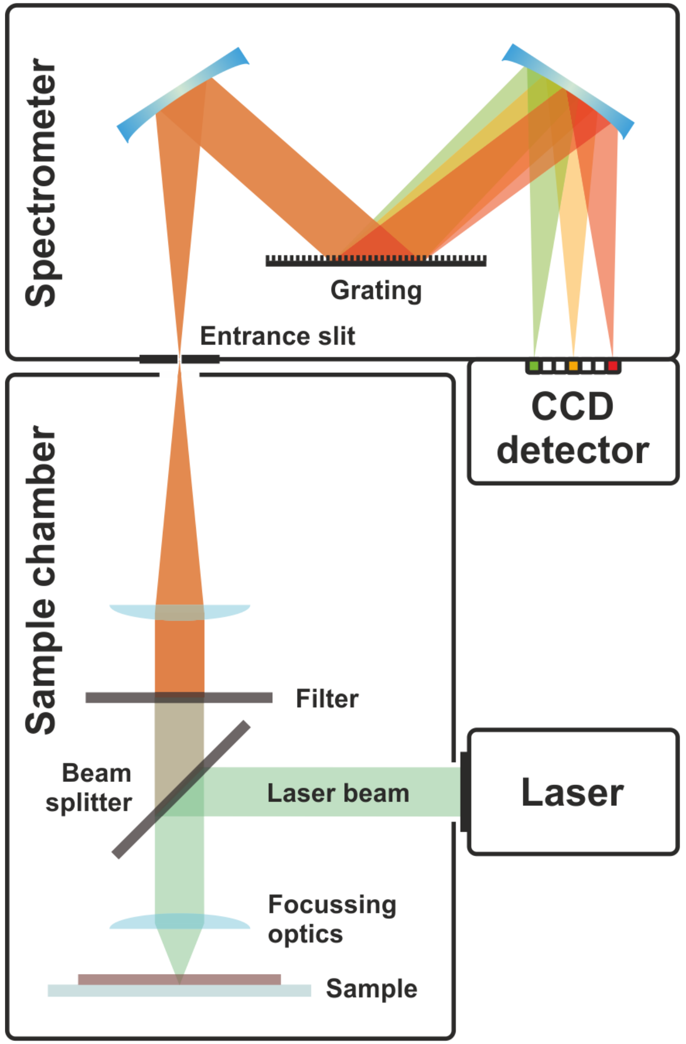
The amplitude is the **Raman matrix element**:

$$M_{if} \sim \langle f|\hat{R}|i\rangle$$

where $\hat{R}$ is the **Raman operator**,

$$\hat{R} = \sum_{mn} \frac{\langle m|\hat{H}_{\mathrm{int}}|n\rangle\langle n|\hat{H}_{\mathrm{int}}|i\rangle}{\omega_{\mathrm{in}} - (E_n - E_i)}.$$

Raman scattering: experimental implementation



Source: Wikipedia

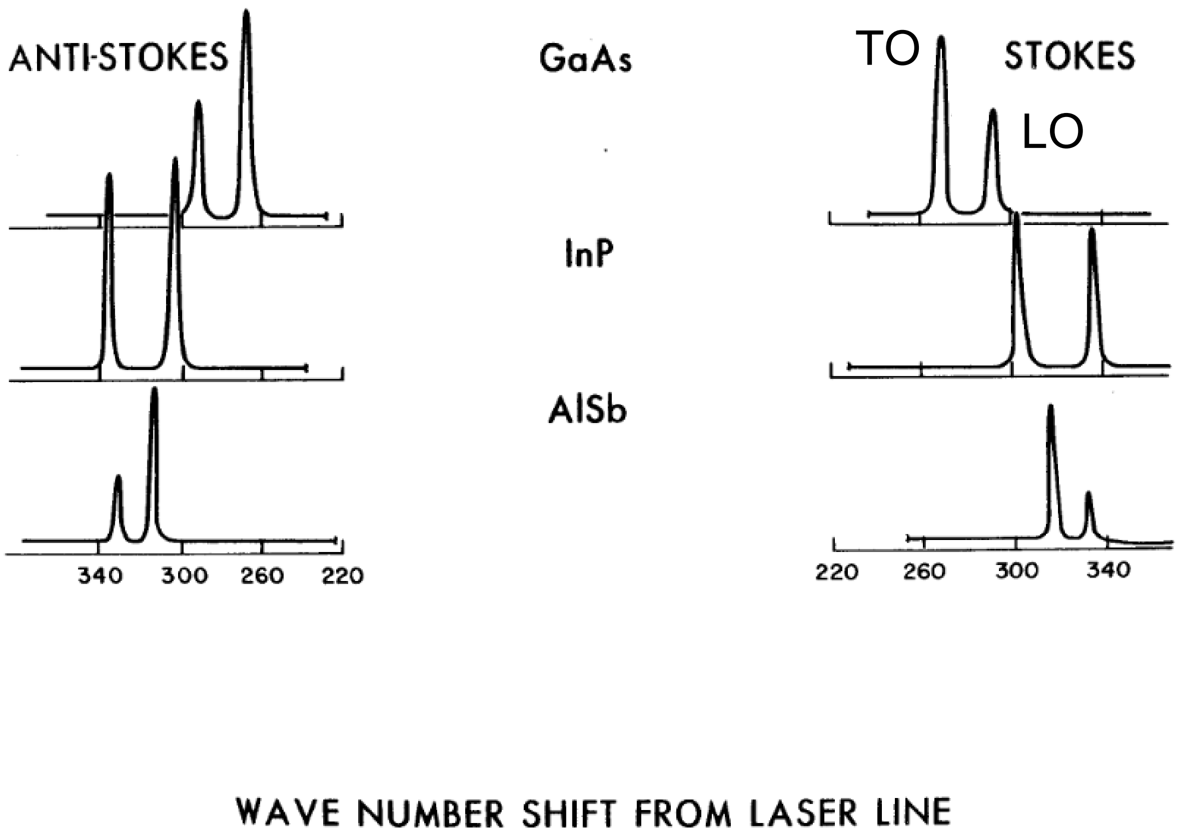
Important considerations

- Raman scattering a weak process → the Rayleigh scattered laser light must by suppressed by many orders of magnitude
- Photons have very little momentum/wavevektor: $k = 2\pi/\lambda$
  → $k \sim 10^{7}\ m^{-1}$
- Wavevectors in Brillouin zone: $k = \pi/a$
  → $k \sim 10^{10}\ m^{-1}$

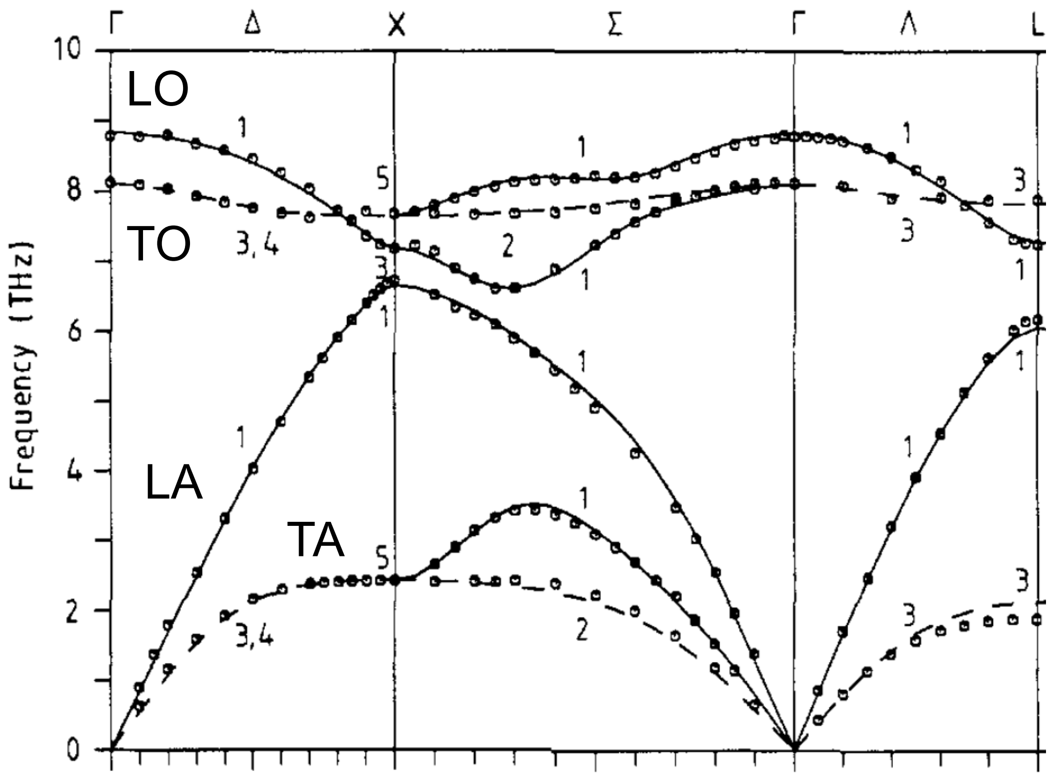
First order Raman scattering probes excitations at q = 0

Raman spectrum of zincblende III-V semiconductors

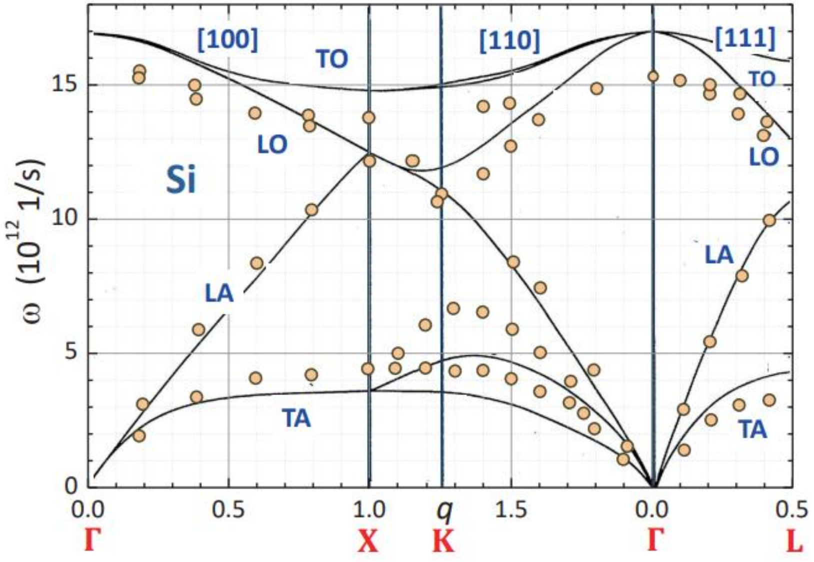
Mooradian and Wright, Solid State Communications Vol. 4, pp.431 - 434, 1966

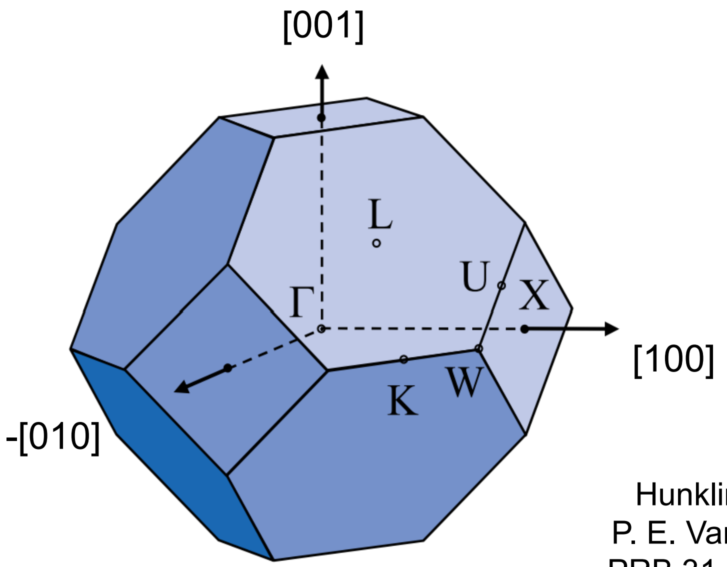


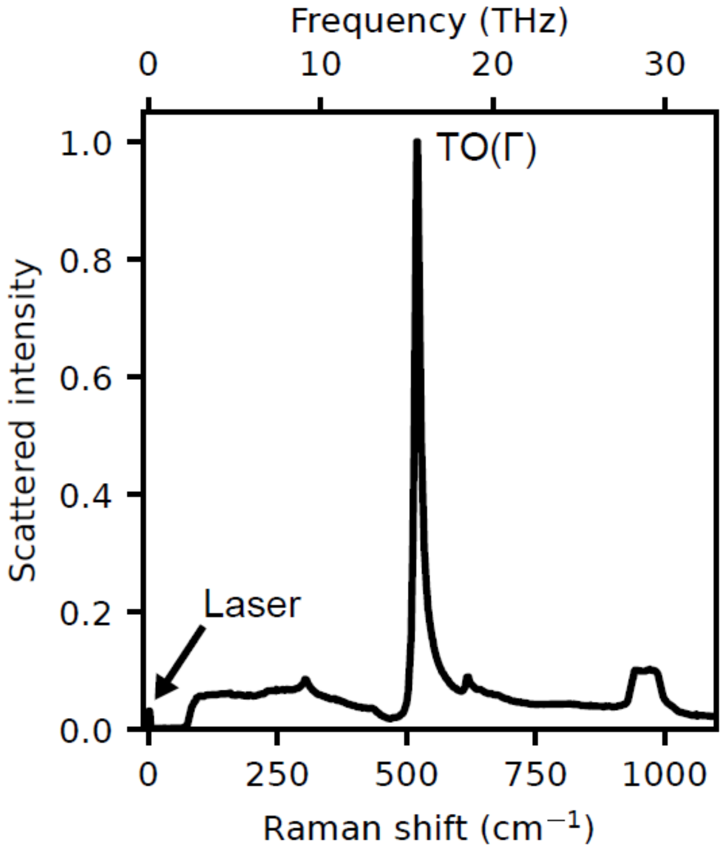
GaAs phonon dispersion

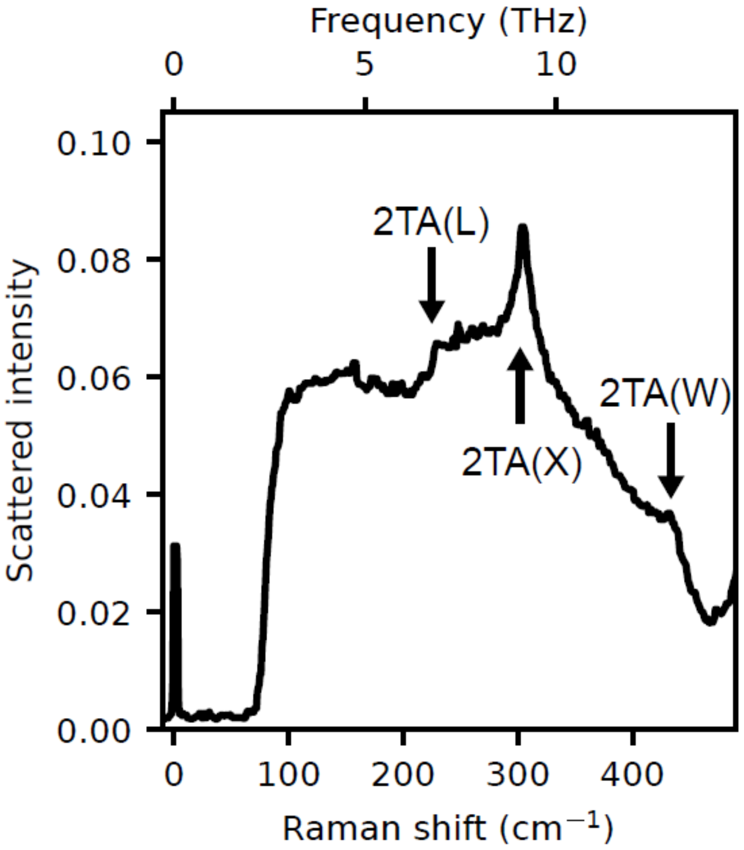


Raman spectrum of silicon





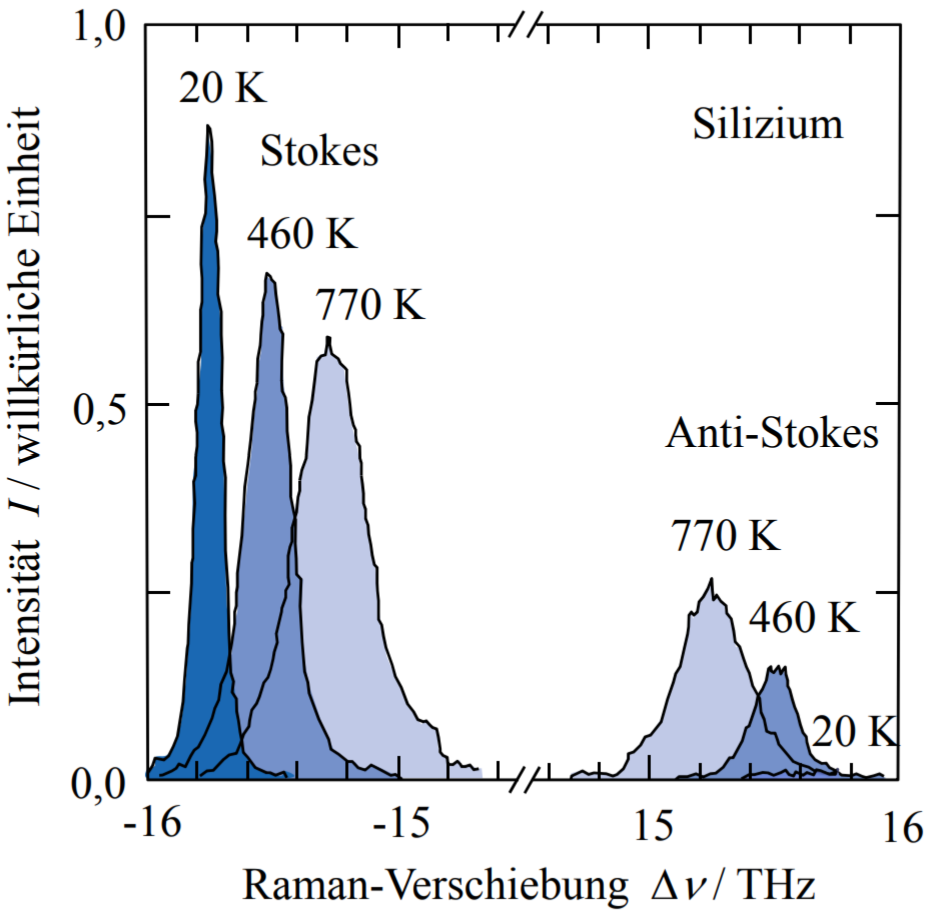




Hunklinger, Gross
P. E. Van Camp et al.,
PRB 31, 4089 (1985).

Raman spectrum of silicon: Stokes vs. Anti-Stokes

Raman-Streuung bzw. Spektroskopie



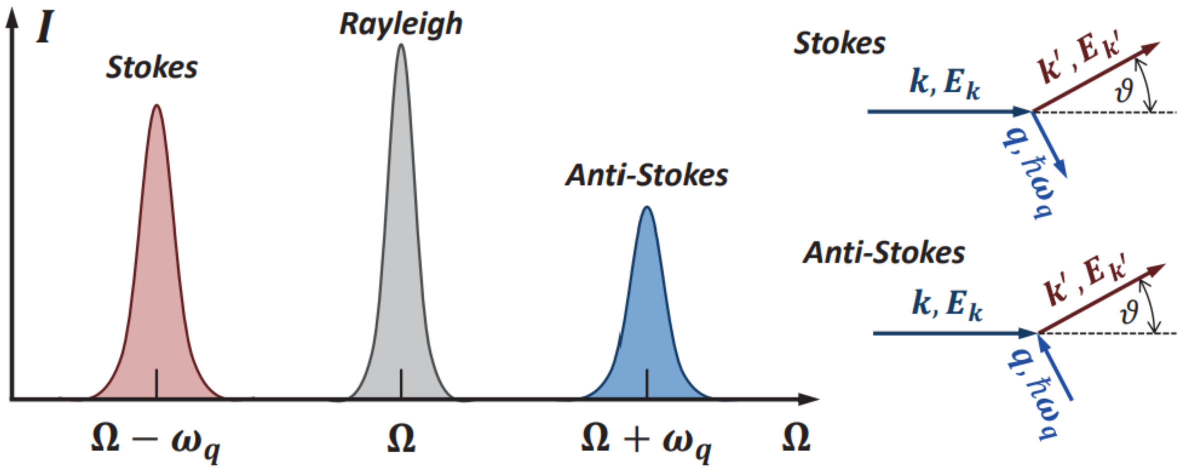


Bild 6.24: Raman-Spektrum von Silizium bei unterschiedlichen Temperaturen. Der Nullpunkt der Frequenzverschiebung ist unterdrückt. (Nach T.R. Hart et al., Phys. Rev. B **1**, 638 (1970).)

aus Hunklinger (Bild 6.24)
& Gross/Marx (Bild 5.21)

Isotope effects

Raman-Streuung bzw. Spektroskopie

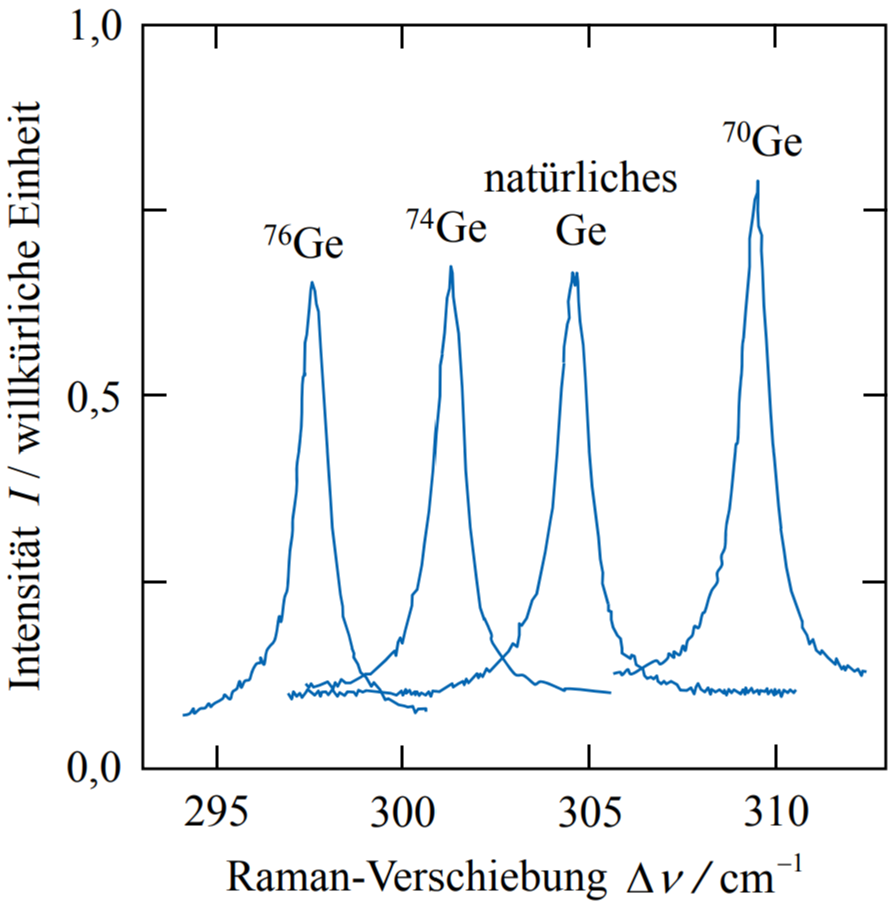


Bild 6.23: Raman-Spektren von isotopenreinem und natürlichem Germanium bei 80 K. Die Verschiebung der Raman-Linie ist in guter Näherung proportional zu $M^{-1/2}$. Die unverschobene Rayleigh-Linie ist nicht zu sehen, da der Nullpunkt der Frequenzverschiebung nicht abgebildet ist. (Nach T. Ruf et al., Phys. Bl. **52**, 1115 (1996).)

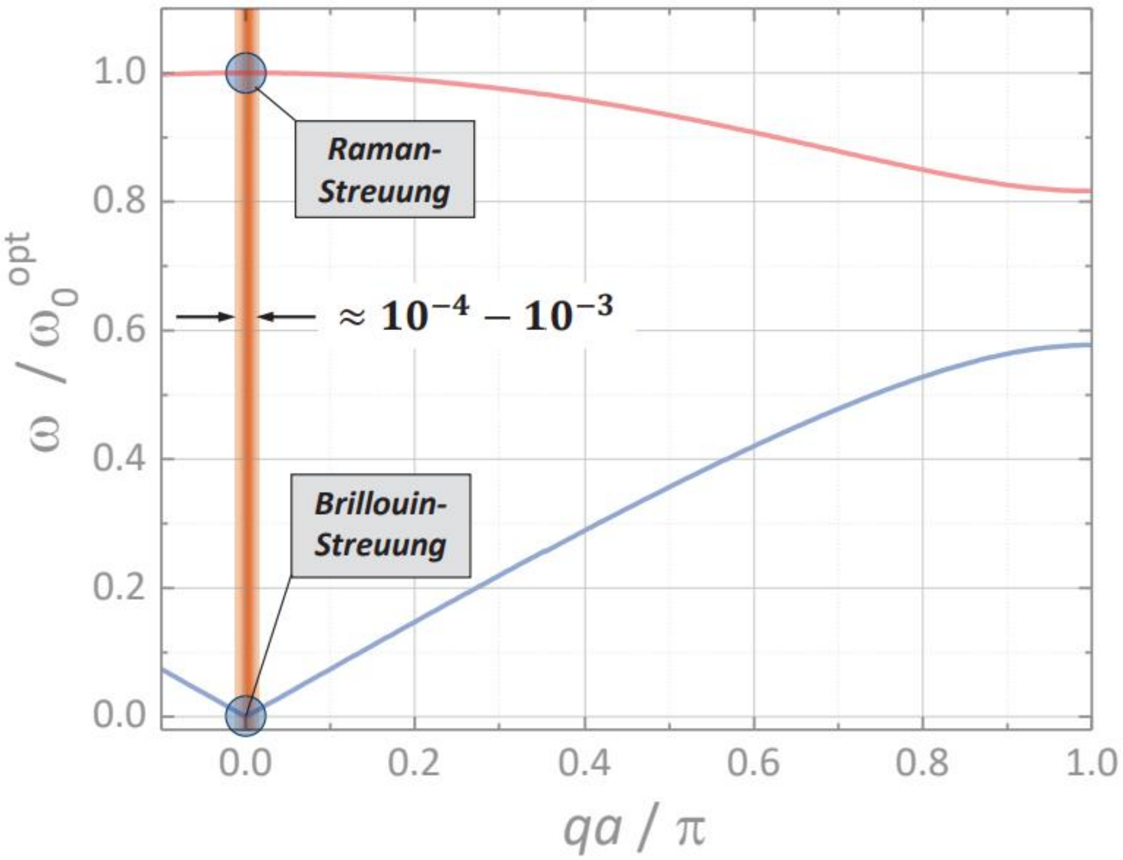
aus Marx/Gross

Sunmary

- Phonons are quasiparticles with well defined energy E and wavevector q
- Occupation number for a mode follows the Bose-Einstein-distribution
- Light-matter interaction is governed by $\boldsymbol{d} \cdot \boldsymbol{E}$ in the dipole approximation
- Raman scattering
    - is formally a second order perturbation theory process
    - probes phonons at the zone center
    - allows Stokes and anti-Stokes processess
    - couples to first order to LO and TO phonons

Brillouin light scattering

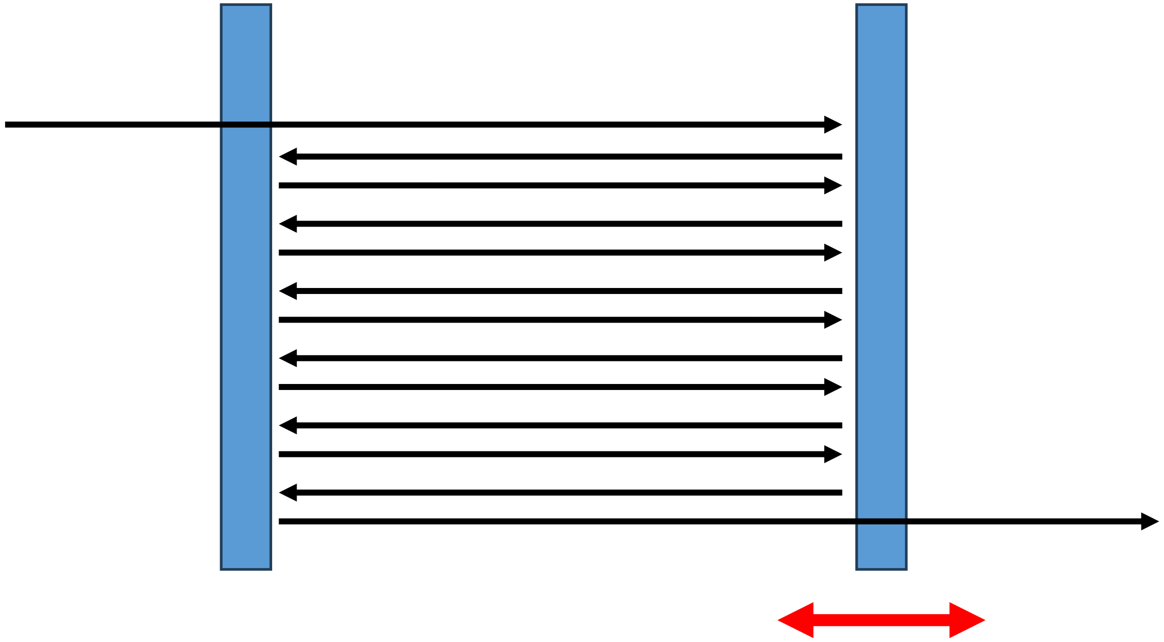
Brillouin- vs. Raman-Streuung



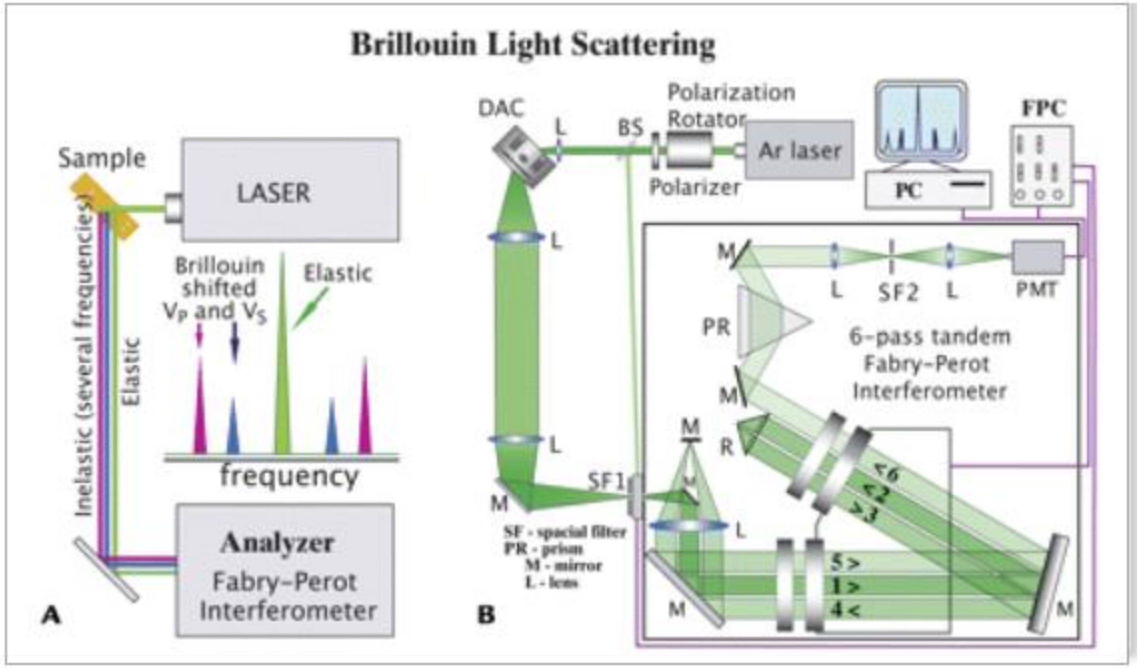
aus Marx/Gross

Brillouin light scattering

Cavity (standing wave)



Movable mirror



Brillouin light scattering

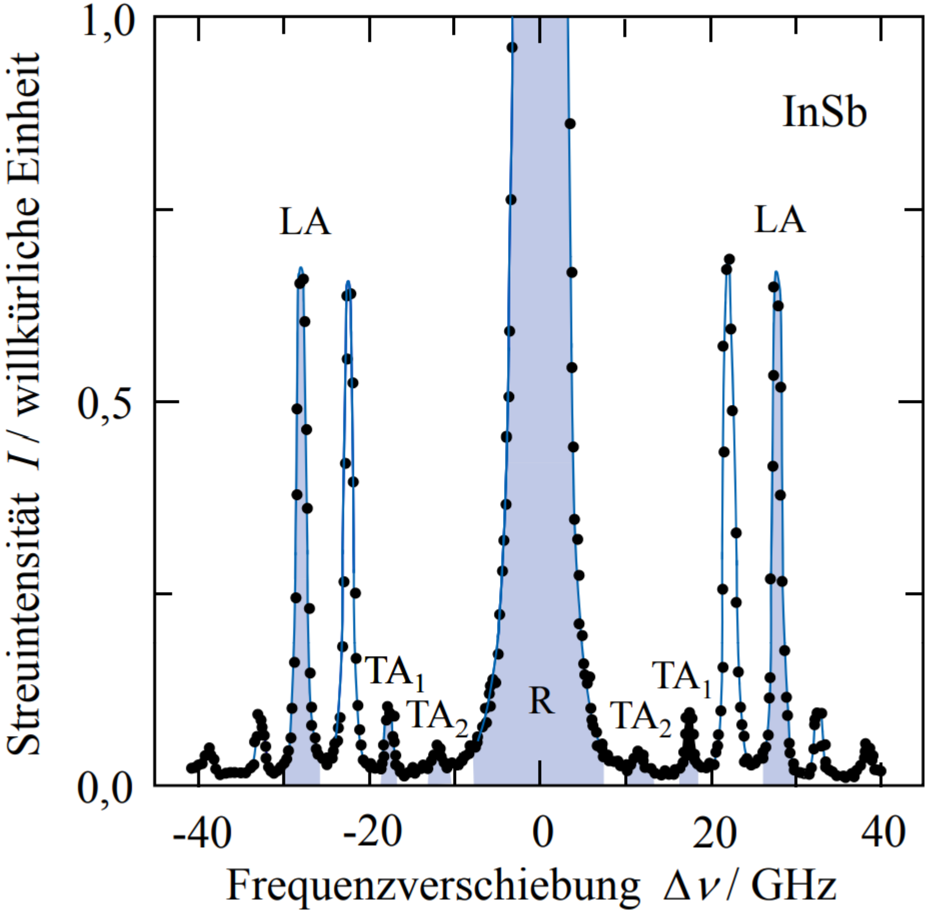


Bild 6.25: Brillouin-Spektrum von Indiumantimonid. Die Messwerte, die zu einer Ordnung gehören, sind hellblau markiert und mit LA, TA_1 und TA_2 bezeichnet. Die dominierende zentrale Linie R rührt von der Rayleigh-Streuung her. (Nach J.R. Sandercock, *Proc.* 2nd *Int. Conf. Light Scattering in Solids*, M. Balkanski, ed., Flammarion Paris, 1971)

aus Hunklinger